

CHROMOSOME KARYOTYPE DATASET

Report

Source: Kaggle — Chromosome Karyotype Images Dataset

1. Introduction

Karyotyping is a clinical diagnostic technique in which human chromosomes are examined under a microscope for abnormalities in number, size, and structure. This dataset has been specifically designed to support the development of deep learning models for automated chromosome classification and abnormality detection — tasks that traditionally require expert cytologists and significant manual effort.

A normal human cell contains 46 chromosomes arranged in 23 pairs. These are grouped into 24 distinct classes for the purpose of classification: chromosomes 1 through 22 (autosomes) and the two sex chromosomes X and Y. The goal of this project is to build a deep learning model that takes a full karyogram image as input and outputs the count of chromosomes per class, while also identifying any structural or numerical abnormalities.

2. Dataset Overview

Dataset Name	Chromosome Karyotype Images
Source	Kaggle (aliabedimadiseh/chromosome-image-dataset-karyotype)
Total Classification Classes	24 (Chromosomes 1–22, X, Y)
Total Images (single_chromosomes_object)	2986 individual chromosome images
Image Type	Grayscale, G-banding technique
Image Dimensions	224 × 224 pixels
Pixel Depth	8-bit (values 0–255, black background)
Annotation Format	CSV files (normal, number_abnormalities, structural_abnormalities)
Dataset Split	train.txt and test.txt text files provided
Release Date	May 2, 2023

3. Dataset Folder and File Structure

The downloaded dataset contains the following folders and files at the root level:

1) Folder: 24_chromosomes_object

This folder contains full karyogram images. Each image in this folder represents a complete chromosome spread from a single patient, where all chromosomes from all 24 classes are present together in one image — exactly as captured from a microscope slide. This is the raw, unsegmented input that would be given to a diagnostic system.

- One image = one patient's complete chromosome spread
- All 24 chromosome classes are present and mixed within a single image
- Chromosomes appear in their natural, scattered arrangement on the microscope slide
- This folder serves as the PRIMARY INPUT to the deep learning pipeline
- Segmentation must be applied to this folder first to isolate individual chromosomes

Think of this as: what the doctor hands the AI — a raw karyogram photo that needs to be analyzed.

2) Folder: single_chromosomes_object

This folder contains pre-segmented, individually cropped chromosome images. Each image contains exactly one chromosome, already cut out and isolated from the full karyogram. These images are labeled by class (1–24) and are used to train the classification model to recognize each chromosome type.

- One image = exactly one isolated chromosome
- Images are labeled by chromosome class (1 through 24)
- Used for TRAINING the classification deep learning model
- Images are grayscale, (224x224) pixels with black background
- The chromosome appears as shades of gray in the center of the image

Think of this as: the labeled flashcards the AI studies to learn what each chromosome looks like.

3) Folder: weight

This folder contains pretrained model weight files from prior training runs on this dataset. These weights can be loaded into a compatible neural network architecture to skip the full training process, or used as a starting point for transfer learning / fine-tuning.

- Contains .pth or .h5 weight files (PyTorch or Keras format)
- Can be used to initialize the model and save training time
- Useful for benchmarking — compare your model accuracy against the pretrained baseline

4) File: normal.csv

A CSV (comma-separated values) file listing all karyogram images that have been confirmed as completely normal. Each row in this file corresponds to one patient image where all 46 chromosomes are present in the correct number and show no structural defects.

- Contains image filenames of normal karyograms
- Used to label the 'healthy' ground truth during training
- A patient listed here should have exactly 2 copies of each autosome class and correct sex chromosomes
- File size: 13 KB, uncompressed ~82 KB — indicating a relatively large number of normal cases

5) File: number_abnormalities.csv

This CSV file lists karyogram images in which the chromosome COUNT is abnormal. A numerical abnormality means a patient has more or fewer chromosomes than expected in one or more classes. These are some of the most clinically significant findings.

- Trisomy: 3 copies of a chromosome instead of 2 (e.g., Trisomy 21 = Down syndrome)
- Monosomy: only 1 copy of a chromosome instead of 2 (e.g., Monosomy X = Turner syndrome)
- Used by the model to flag when a predicted count per class deviates from the expected value
- File size: 1 KB — numerical abnormalities are rarer than normal cases in this dataset

6) File: structural_abnormalities.csv

This CSV file lists images in which chromosomes are present in the correct number but have abnormal structure. A structural abnormality means part of a chromosome is missing, duplicated, flipped, or has fused with another chromosome. The chromosome belongs to the correct class but does not look normal.

- Deletion: a portion of a chromosome arm is missing — chromosome looks shorter than normal
- Translocation: a fragment from one chromosome attaches to another chromosome
- Inversion: a segment of the chromosome is reversed — banding pattern appears flipped
- Duplication: a section is copied and inserted again — chromosome appears longer
- Ring chromosome: both ends fuse into a ring shape — visually very distinctive
- Used to train the abnormality detection component of the model

7) File: train.txt

A plain text file listing the filenames of all images designated for model training. The model is trained exclusively on the images listed here. This split ensures consistency and reproducibility across different experiments.

- File size: 8 KB uncompressed (~33 KB) — the larger split, containing the majority of images
- Contains one image filename per line
- Used with the single_chromosomes_object folder to load training batches
- Typical split ratio: ~80% of total images assigned to training

8) File: test.txt

A plain text file listing the filenames of all images reserved for testing and evaluating the model after training. These images are never seen by the model during training — they simulate real-world unseen data.

- File size: 3 KB (~11 KB uncompressed) — smaller split, approximately 20% of images
- Used to measure final model accuracy, precision, recall, and F1 score
- Images in this file must NEVER be used during training to avoid data leakage

9) File: diff_image.txt

A plain text file that lists images classified as 'difficult' — these are images that are harder for the model to classify correctly. Difficult images may include chromosomes that are bent, overlapping, blurry, or poorly stained.

- Helps researchers understand where the model is likely to fail
- Can be used to apply extra data augmentation on hard samples
- Useful for building a more clinically robust model that handles edge cases

10) File: Readme.txt

A plain text documentation file provided by the dataset authors describing the dataset contents, collection methodology, and usage guidelines. Should be read before using the dataset.

4. Image Count Per Class — single_chromosomes_object Folder

The table below lists the exact number of images available for each chromosome class in the single_chromosomes_object folder. These counts are based on published research using this dataset (Lin et al., CIR-Net). The dataset contains 2,986 total individual chromosome images across all 24 classes.

Class #	Chromosome	Denver Group	Expected per Person	Images in Dataset	Notes
1	Chromosome 1	A – Large metacentric	2	130	—
2	Chromosome 2	A – Large submetacentric	2	130	—
3	Chromosome 3	A – Large metacentric	2	130	—
4	Chromosome 4	B – Large submetacentric	2	130	—
5	Chromosome 5	B – Large	2	130	—

Class #	Chromosome	Denver Group	Expected per Person	Images in Dataset	Notes
		submetacentric			
6	Chromosome 6	C – Medium submetacentric	2	130	—
7	Chromosome 7	C – Medium submetacentric	2	130	—
8	Chromosome 8	C – Medium submetacentric	2	129	—
9	Chromosome 9	C – Medium submetacentric	2	130	—
10	Chromosome 10	C – Medium submetacentric	2	130	—
11	Chromosome 11	C – Medium submetacentric	2	130	—
12	Chromosome 12	C – Medium submetacentric	2	129	—
13	Chromosome 13	D – Medium acrocentric	2	130	—
14	Chromosome 14	D – Medium acrocentric	2	130	—
15	Chromosome 15	D – Medium acrocentric	2	130	—
16	Chromosome 16	E – Small metacentric	2	129	—
17	Chromosome 17	E – Small submetacentric	2	130	—
18	Chromosome 18	E – Small submetacentric	2	130	—
19	Chromosome 19	F – Small metacentric	2	129	—
20	Chromosome 20	F – Small metacentric	2	130	—
21	Chromosome 21	G – Smallest acrocentric	2	130	—
22	Chromosome 22	G – Smallest acrocentric	2	130	—

Class #	Chromosome	Denver Group	Expected per Person	Images in Dataset	Notes
23	X Chromosome	Sex chromosome	1 (male) / 2 (female)	98	Imbalanced
24	Y Chromosome	Sex chromosome	0 (female) / 1 (male)	32	Imbalanced
TOTAL	24 Classes	Groups A–G + Sex	46 (typical)	2986	

Important note: Classes 8, 12, 16, and 19 contain 129 images instead of 130 due to minor data collection gaps. Classes 23 (X) and 24 (Y) are significantly underrepresented — X has 98 images and Y has only 32 images. This class imbalance must be addressed during model training using techniques such as weighted loss functions or data augmentation.

5. Model Input and Output

1) Input Features

The model receives the following as input:

- A full karyogram image from the 24_chromosomes_object folder
- Image format: Grayscale JPEG/PNG, 224 x 224 pixels (after resizing)
- Pixel values: 0 (black background) to 255 (bright chromosome bands)
- The G-banding pattern — alternating dark and light bands on each chromosome — is the key visual feature the model learns to differentiate between classes
- No manual feature extraction needed — the CNN automatically learns shape, length, centromere position, and banding pattern

2) Expected Output

For each input karyogram image, the model should produce:

- The total count of chromosomes detected per class (1 through 24)
- Whether each detected chromosome is structurally normal or abnormal
- A summary of total chromosomes found and whether the karyotype is normal or abnormal

Example output format for one patient image:

Class	Chromosome	Found (Count)	Normal	Abnormal / Flag
1	Chr 1	2	2	0

Class	Chromosome	Found (Count)	Normal	Abnormal / Flag
2	Chr 2	2	1	1 (deletion detected)
3	Chr 3	2	2	0
4	Ch4	2	2	0
5	Ch 5	2	2	0
6	Ch 6	2	2	0
7	Ch 7	2	2	0
8	Ch 8	2	2	0
9	Ch 9	2	2	0
10	Ch 10	2	2	0
11	Ch 11	2	2	0
12	Ch 12	2	2	0
13	Ch 13	2	2	0
14	Ch 14	2	2	0
15	Ch 15	2	2	0
16	Ch 16	2	2	0
17	Ch 17	2	2	0
18	Ch 18	2	2	0
19	Ch 19	2	2	0
20	Ch 20	2	2	0

21	Chr 21	3	2	1 (Trisomy — extra copy!)
22	Chr 22	2	2	0
23	X	2	2	0
24	Y	0	0	0
TOTAL	—	47 found	—	Abnormal karyotype — Classes 2, 21 flagged

6. Project Pipeline

The complete deep learning pipeline for this project is as follows:

Step	Stage	Description
1	Input	Receive full karyogram image from 24_chromosomes_object
2	Segmentation	Detect and crop individual chromosomes from the full image using a segmentation model (U-Net / Mask R-CNN)
3	Classification	Classify each cropped chromosome into one of 24 classes using a CNN trained on single_chromosomes_object images
4	Counting	Count the number of chromosomes detected per class
5	Abnormality Detection	Compare each chromosome against the normal reference from normal.csv; flag numerical and structural abnormalities
6	Report Generation	Output a class-by-class summary with counts and abnormality flags

7. Key Challenges and Considerations

- **Y chromosome (class 24) has only 32 images compared to 130 for most other classes. Weighted loss or oversampling required:** Class imbalance
- **In 24_chromosomes_object images, chromosomes may overlap or touch, making segmentation difficult. The diff_image.txt file identifies the hardest cases:** Overlapping chromosomes
- **Chromosomes are non-rigid and can appear bent or curved, making classification harder. Data augmentation (rotation, flipping) is needed:** Chromosome bending
- **Staining intensity can vary between samples, affecting the appearance of banding patterns across different images:** G-banding staining variation
- **These are two separate detection tasks requiring different model components or multi-task learning:** Structural vs numerical abnormalities

8. Recommended Model Architecture

Task	Recommended Model	Reason
Segmentation	U-Net or Mask R-CNN	Proven for biomedical image segmentation
Classification (24 classes)	ResNet-50 or EfficientNet-B3	Strong feature extraction, high accuracy on small datasets
Abnormality detection	Multi-task head on classifier	Shares features with classifier, lightweight addition
Framework	PyTorch or TensorFlow/Keras	Both support all the above architectures
Preprocessing	OpenCV + scikit-image	Image resizing, normalization, augmentation

References

1. Lin, C. et al. (2023). CIR-Net: Chromosome Image Recognition Network. GitHub Repository. <https://github.com/CloudDataLab/CIR-Net>
2. Scientific Data — An Open Dataset of Annotated Metaphase Cell Images for Chromosome Identification. Nature Publishing Group, February 2023.
3. Deep-Learning-Based Human Chromosome Classification: Data Augmentation and Ensemble. MDPI Information, July 2023.
4. Kaggle Dataset — Chromosome Karyotype Images. <https://www.kaggle.com/datasets/aliabedimadiseh/chromosome-image-dataset-karyotype>